Who Cares If The Computer Breaks?

MIT Forum — October 1995

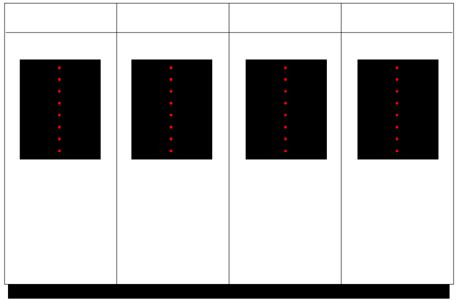
Peter S. Kastner
Vice President
Aberdeen Group, Inc.
Boston, Mass. 02108
(617) 723-7890

Agenda

- Who cares if the computer breaks?
- Where computer systems fail
- Planning for the inevitable
- Approaches to minimizing downtime

Who Cares If the Computer Breaks?

Computer Industry Ancient History



- One-tier computing
  - Jobs were batch-processed based on keypunched data entered on forms by people
  - Reports printed on paper
  - Virtually no one was on-line
- Commercial computers kept the books
- If the computer failed, jobs were restarted and run again

Fast Forward to the 1990s

- Complex computing environments
  - Multiple tiers of networks
  - Computer-to-computer transactions
  - On-line, real-time changes
  - Less paper, ad hoc queries
- Commercial computers *make* the money
- But computer systems still fail!

Who Cares If The Computer Breaks?

- Public safety
- National security
- Banking and financial system
- Telecommunications industry
- Retailers
- 45% of U.S. workers now using PCs
- And virtually every aspect of our society

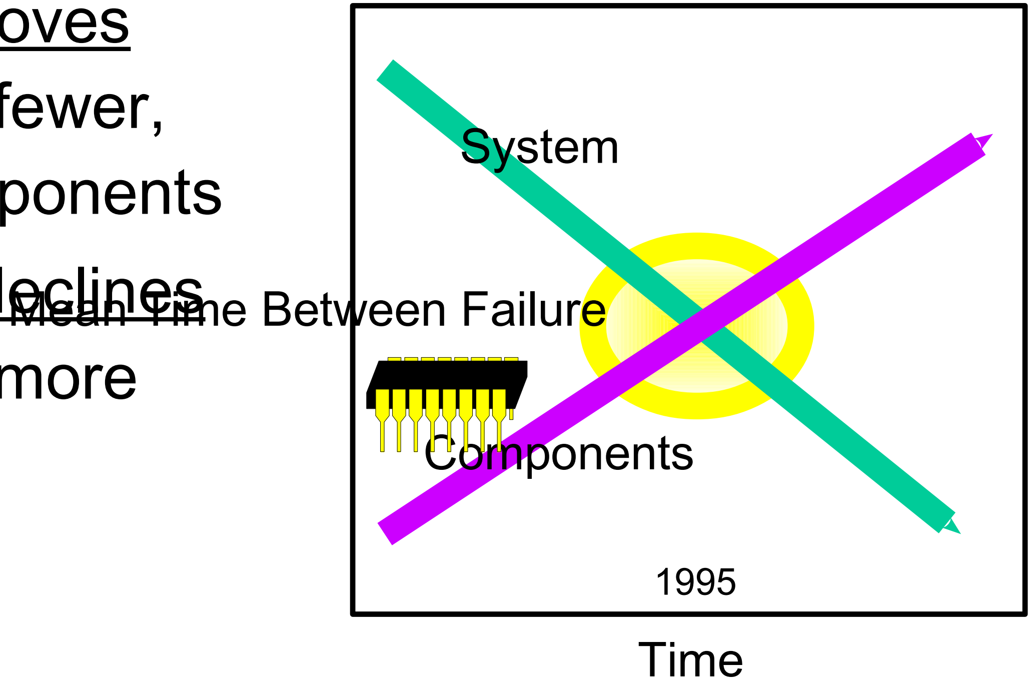
Where Computer Systems Fail

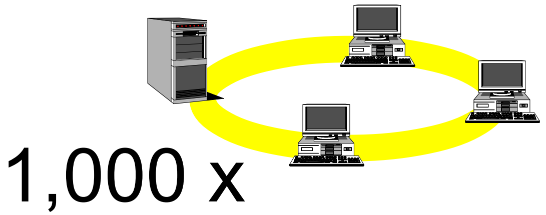
Hardware, Networks, & Software

Hardware Always Breaks

- Unit reliability <u>improves</u> over time due to fewer, more robust components
- System reliability <u>declines</u> over time due to more components

Reliability Trends





Software Usually Breaks

- Correct software will outlive its hardware and never fail
- But most software fails
  - Too many lines of code ... complexity exceeds human capacity
  - Poor or unsystematic design, review, testing
  - Poor choice of software development tools
  - Repeated re-work due to business logic changes
  - Trade-offs on quality vs. time-to-market

Networks Will Always Break

- Networks suffer all the inherent problems of computer hardware and software, plus the reliability problems of electrical/light cables
- Network availability is a key inhibitor to the growth of distributed computing

Planning for the Inevitable

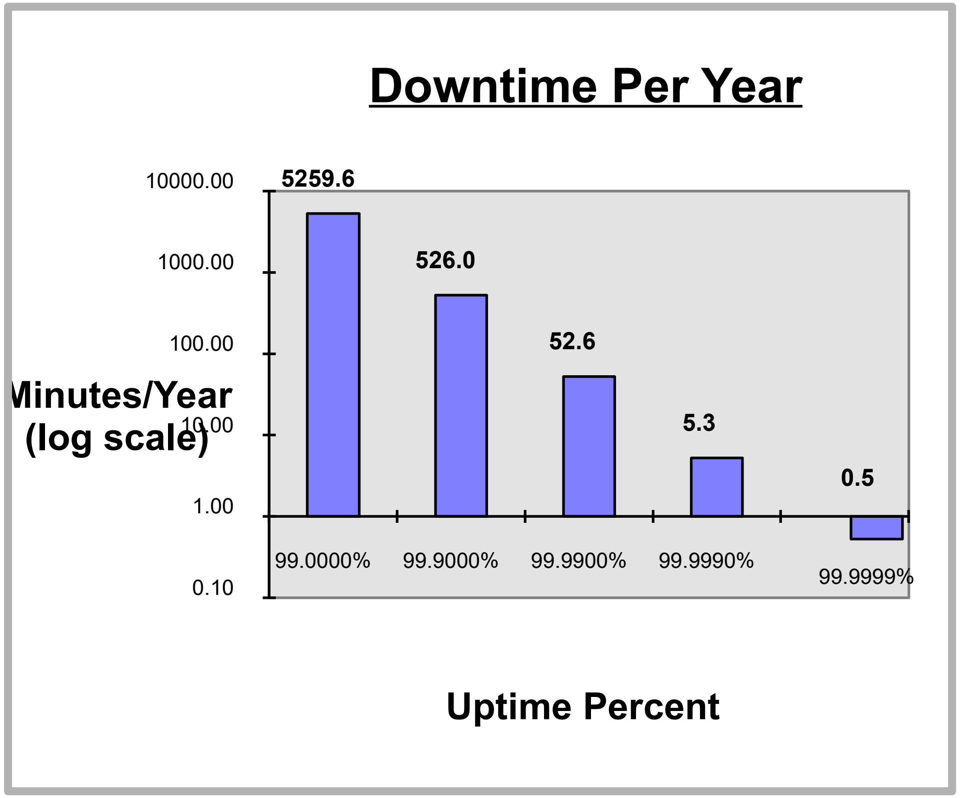
“A stitch in time saves nine”

Planning for Downtime

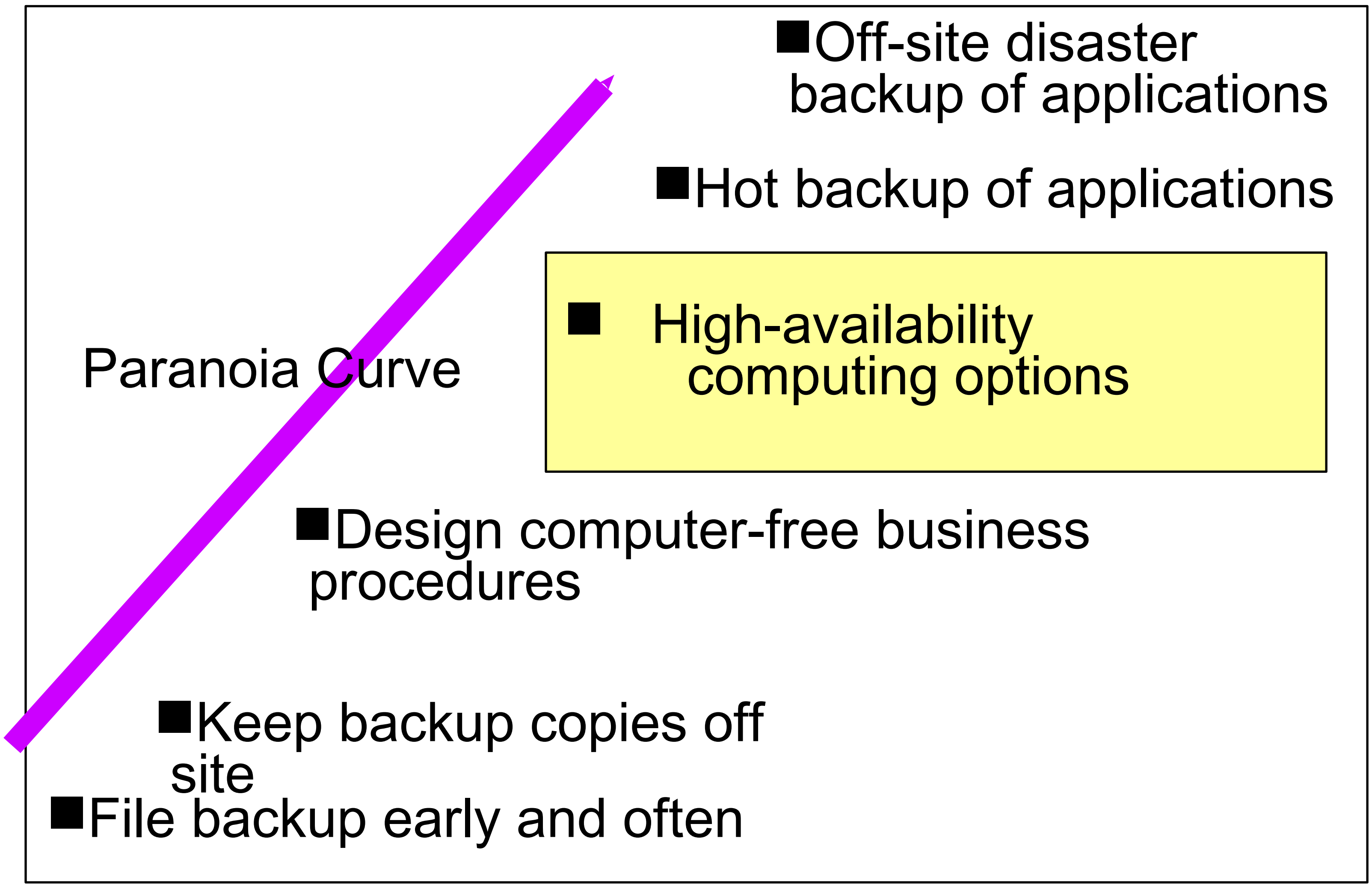
- Backup files early and often
- Keep backup copies off site
- Design computer-free business procedures
- Choose high-availability computing options
- Hot backup of applications
- Off-site disaster backup of applications

Application Availability

- There are lots of ways to measure availability
  - MTBSI
  - Hardware availability
  - System availability
- Application availability to users is the most useful measure
- In today's business world, 99.9% is a floor, not the ceiling



Planning for Downtime



Approaches to Minimizing Downtime

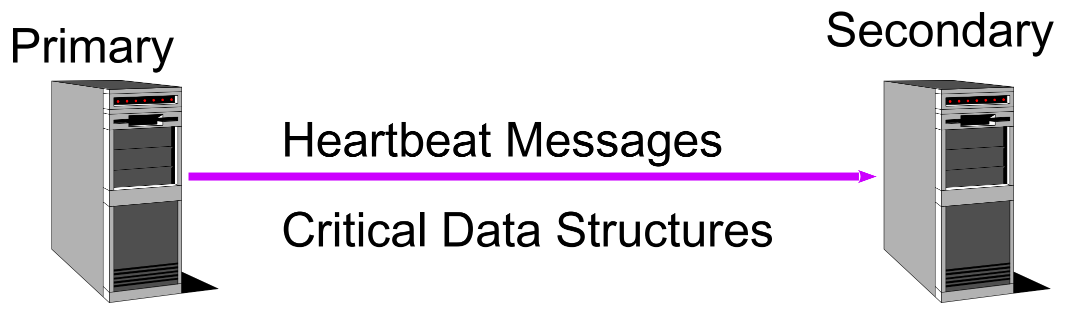
- *How paranoid are you?*
- *How much have you got to spend?*
  - Hardware
  - Software
  - Networks
  - Other

Hardware

- Standard industry practices to catch errors
- Self-testing (including systems management software)
- Overcapacity by design
- Hot replacement of power, fan, disk components
- RAID disk storage options
- Redundant hardware with failover (N+1 or other)

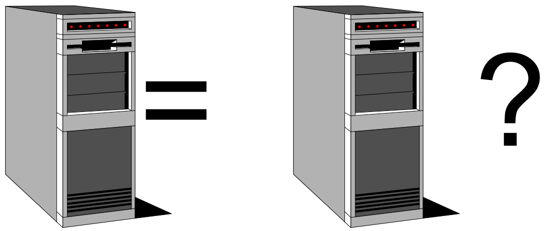
Classic Failover Architecture

- **Primary Machine**
  - Runs critical application
  - Saves key data periodically
  - Sends periodic "heartbeat" message to secondary
  - Eventually fails ...
- **Secondary Machine**
  - Listens for periodic heartbeat message
  - Determines when Primary has stopped executing
  - Uses saved key data to restart/recover application



Hardware Failover Variations

Hardware Self-Checking

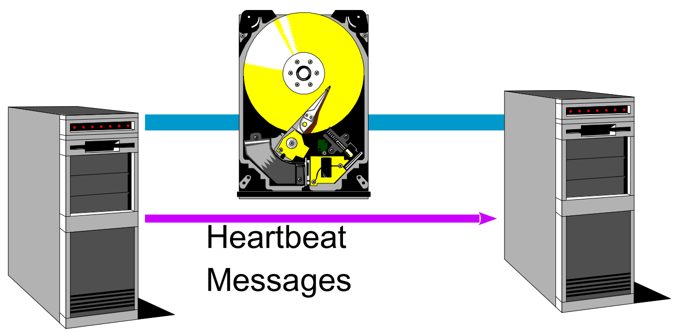


Stratus

N + 1

Tandem

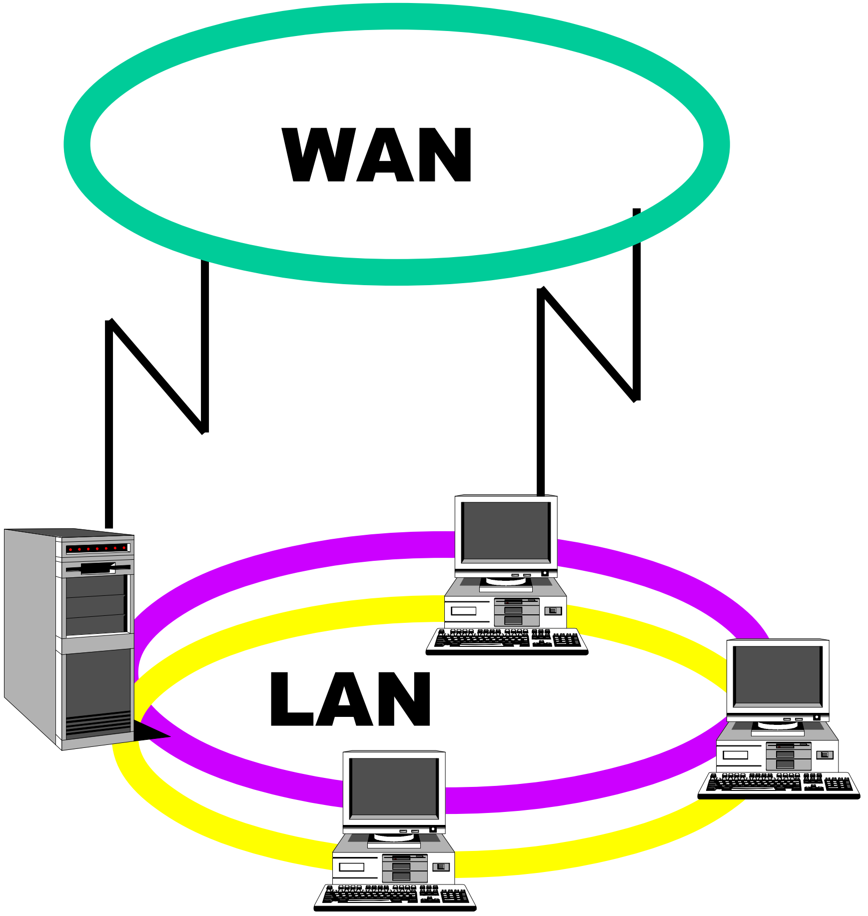
Shared Disk



Digital Equipment

Networks

- Redundant network interface hardware
- Bridge/routers follow computer hardware path
- Robust protocols
- Adaptive routing eliminates point-point failure
- More options in high-end servers; few options in workgroup servers



Software Availability

- Failover always involves software
- Failure prediction technology aims to catch hardware failures before they happen
- Operating system availability is directly related to age
- RDBMS & OLTP software ensures data recoverability
- Applications are poorly instrumented for failures, but will improve
- Desktop software is still prone to failure

Opportunities

- Hardware and Networks
  - Note: buyers are contradictory. They want the highest availability levels but are unwilling to pay (much) more than commodity prices
  - Low/no-cost improvements in client-server application availability
  - Added value by improved failure prediction due to better error-recovery reporting
  - Self-repairing networks

Software and Services

- Software
  - –Systems management tools down to the application-object level
  - –Adaptive routing to movable objects
  - –Client-server application testing tools
  - –Self-distributing databases
- Services
  - –Disaster backup for client-server applications
  - –Availability audits and testing

Aberdeen Group Conclusions

- Computer failures are becoming more catastrophic because we depend so much on computers
- Reducing the cost and aggravation of downtime involves a spectrum of increasingly costly remedies
- Hardware and network availability improvements are offset by more computers/networks that will fail
- Most future improvements in availability will depend on system software that assists application failover without dramatically increasing development costs or complexity
- There are interesting opportunities for new businesses

ABERDEEN GROUP • CHARTING THE COURSE